FINAL EXAMINATION

December 13, 2025

Question	Points	Score
Basic Concepts	55	
Gradient-Based Optimization of ML Models	10	
Neural Network Construction and Analysis	10	
Training vs. Testing	10	
Transformers and Large Language Models	15	
Total:	100	

- Please write down your **name** and **student ID** on the **answer paper**.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

Question 1 [55 points]: Basic Concepts

The instructions are as follows:

- **Single-Choice Questions:** There is exactly one correct answer.
- **Multiple-Choice Questions (worth 5 pts):** There are at least two correct answers. The grading policy is as follows:
  - You receive 5 points only if you select **all** correct options and **no** incorrect options.
  - Missing a correct option deducts **2 pts**; selecting an incorrect option deducts **2 pts**. Missing two correct options deducts **5 pts**; selecting two incorrect options deducts **5 pts**. Simply selecting all options deducts **5 pts** (no question will have all options correct).
- You *do not* need to justify your answer in Question 1.

(a) [3 points] In the Perceptron algorithm, if the current model parameters are $\boldsymbol{\theta}$ and the algorithm misclassifies a training point $(\boldsymbol{x}_i, y_i)$ where $y_i \in \{-1, +1\}$, the update rule is $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - y_i \boldsymbol{x}_i$. (true or false)

(b) [3 points] Which one of the following conditions guarantees a unique closed-form solution $\widehat{\boldsymbol{\theta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$ for the Least Squares problem $\min_{\boldsymbol{\theta}} \frac{1}{n} \|\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\|_2^2$?

(1) The number of samples n is less than the number of features d.
(2) The matrix $\boldsymbol{X}$ has full column rank.
(3) The matrix $\boldsymbol{X}$ is a square matrix.
(4) The target values $\boldsymbol{y}$ are all positive.

(c) [3 points] According to the generalization bound for a finite hypothesis space $\mathcal{H}$, how does the generalization error bound scale with the size of the hypothesis space $|\mathcal{H}|$?

(1) Linearly with $|\mathcal{H}|$.
(2) Linearly with $\sqrt{|\mathcal{H}|}$.
(3) Logarithmically, i.e., proportional to $\sqrt{\log |\mathcal{H}|}$.
(4) It is independent of $|\mathcal{H}|$.

(d) [5 points] Which of the following strategies help in reducing the generalization error (making $\mathrm{Er}_{\mathrm{out}}$ close to $\mathrm{Er}_{\mathrm{in}}$)? (multiple choice)

(1) Choosing a hypothesis set $\mathcal{H}$ with a smaller VC dimension.
(2) Choosing a hypothesis set $\mathcal{H}$ with a larger growth function $\mathcal{G}_{\mathcal{H}}(n)$.
(3) Increasing the number of training samples n.
(4) Choosing a hypothesis set $\mathcal{H}$ with infinite VC dimension.
(5) Choosing a hypothesis set $\mathcal{H}$ with a larger number of effective parameters.

(e) [3 points] Which one of the following statements correctly describes Nesterov's Accelerated Gradient Descent (AGD) compared to standard Gradient Descent with Momentum?

(1) AGD evaluates the gradient at the current iterate $\boldsymbol{\theta}_k$.
(2) AGD evaluates the gradient at an extrapolated point $\boldsymbol{w}_k$, which is a combination of $\boldsymbol{\theta}_k$ and $\boldsymbol{\theta}_{k-1}$.
(3) AGD has a slower convergence rate of $\mathcal{O}(1/k)$ compared to the $\mathcal{O}(1/k^2)$ rate of standard Gradient Descent.
(4) AGD requires computing the Hessian matrix (second-order derivations) of the objective function.

(f) [3 points] When using a validation set of size K to estimate the out-of-sample error $\mathrm{Er}_{\mathrm{out}}(f)$, the approximation error $|\,\mathrm{Er}_{\mathrm{out}}(f) - \mathrm{Er}_{\mathrm{val}}(f)|$ is bounded by a term proportional to:

(1) $1/K$
(2) $1/\sqrt{K}$

(3) $\sqrt{K}$
(4) e^{-K}

(g) [5 points] Which of the following factors are identified as "catalysts" for overfitting (i.e., increasing them tends to increase overfitting)? (multiple choice)
(1) High noise level (σ^2) in the data.
(2) Large number of training data points (n).
(3) Very high target model complexity.
(4) Large regularization parameter (λ) in weight decay.
(5) Hypothesis space $\mathcal{H}$ with a high VC dimension.

(h) [3 points] For non-linearly separable dataset, the optimization formulation of hard-margin SVM can still be solved. This will return a classifier that correctly classifies the separable data points.

(i) [3 points] Consider the Gaussian/RBF kernel for d-dimensional data $\boldsymbol{x} \in \mathbb{R}^d$. What is the dimension p of the corresponding lifted feature space $\Phi(\boldsymbol{x}) \in \mathbb{R}^p$?

(j) [5 points] Which of the following optimization problems can be used to derive the Principal Component Analysis (PCA) solution? (multiple choice)
(1) $\min_{\boldsymbol{A}, \boldsymbol{A}^\top \boldsymbol{A} = \mathbf{I}} \sum_{i=1}^n \|\boldsymbol{x}_i - \boldsymbol{A}\boldsymbol{A}^\top \boldsymbol{x}_i\|_2^2$.
(2) $\max_{\boldsymbol{A}, \boldsymbol{A}^\top \boldsymbol{A} = \mathbf{I}}$ trace$(\boldsymbol{A}^\top \boldsymbol{S} \boldsymbol{A})$ where $\boldsymbol{S}$ is the empirical covariance matrix.
(3) $\min_{\boldsymbol{A}, \boldsymbol{A}^\top \boldsymbol{A} = \mathbf{I}} \sum_{i=1}^n \|\boldsymbol{A}^\top \boldsymbol{x}_i\|_2^2$.
(4) $\min_{\boldsymbol{A}, \boldsymbol{A}^\top \boldsymbol{A} = \mathbf{I}, \boldsymbol{\Theta}} \|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{\Theta}\|_F^2$.

(k) [3 points] The optimization problem for training a two-layer neural network (with a hidden layer $\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x})$ and output $\boldsymbol{y} = h(\boldsymbol{W}\boldsymbol{z})$) is a convex optimization problem with respect to the weights $\boldsymbol{V}$ and $\boldsymbol{W}$. (true or false)

(l) [3 points] Which of the following statements best describes the tokenization strategy used in contemporary Large Language Models?
(1) They use character-level tokenization to ensure the vocabulary size is as small as possible.
(2) They use whole-word tokenization, assigning a unique ID to every word in the English dictionary.
(3) They use a randomized hashing method to map words to integers without a fixed vocabulary.
(4) None of the above.

(m) [3 points] To finetune an LLM with M billion parameters using the AdamW optimizer and a mixed precision training approach, the approximate amount of GPU RAM required to store the model weights, gradients, and optimizer states (excluding activation memory) is:
(1) $4M$ GB
(2) $12M$ GB
(3) $16M$ GB
(4) $18M$ GB

(n) [5 points] Which of the following statements regarding the Transformer architecture and its normalization methods are correct? (multiple choice)
(1) In a standard Post-Norm Transformer layer, the normalization is applied to the input $\boldsymbol{X}$ before entering the Multi-Head Self-Attention module.
(2) Root Mean Square Normalization (RMSNorm) differs from LayerNorm by using only a scaling parameter $\boldsymbol{\gamma}$ and omitting the shifting parameter $\boldsymbol{\beta}$.
(3) Modern LLMs often prefer Pre-Norm because it provides a more stable training process compared to Post-Norm, and Pre-Norm has the formula ($\boldsymbol{Z} = \boldsymbol{Y}(\text{Norm}(\boldsymbol{X})) + \boldsymbol{X}$).
(4) The Mixture-of-Experts (MoE) transformer architecture replaces the Self-Attention layer with a sparse router.
(5) Flash-Attention reduces the computational complexity of the attention mechanism from $\mathcal{O}(n^2 d)$ to linear time $\mathcal{O}(nd)$.

(o) [5 points] Choose the correct statements regarding the post-training of LLMs. (multiple choice)
  (1) SFT uses a completely different loss function formulation compared to Pre-training.
  (2) In SFT, a loss mask is used to ensure that the "question" or "prompt" tokens do not contribute to the loss calculation.
  (3) Direct Preference Optimization (DPO) minimizes the likelihood of the preferred response $\boldsymbol{y}_w$ while maximizing the likelihood of the rejected response $\boldsymbol{y}_l$, relative to a reference model.
  (4) For mathematical reasoning, pass@k score will not increase when k increases.
  (5) In the RLVR (e.g., GRPO) framework, the advantage A_i for a rollout is calculated using the rewards of the group of sampled outputs.

Question 2 [10 points]: Gradient-Based Optimization of ML Models

Consider the ℓ_2-regularized binary logistic regression problem:

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \left\{ \mathcal{J}(\boldsymbol{\theta}) := \underbrace{\frac{1}{n}\sum_{i=1}^{n} \log\left(1+\exp\left(-y_i\cdot\boldsymbol{\theta}^\top\boldsymbol{x}_i\right)\right)}_{\mathcal{L}(\boldsymbol{\theta})} + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2 \right\}, \quad \lambda > 0.$$

We will explore gradient computation and different optimization algorithms for solving such problems.

(a) [3 points] Compute the gradient of the joint objective function $\mathcal{J}(\boldsymbol{\theta})$.

(b) [3 points] Write down the algorithmic update rules for AdaGrad and AdamW at iteration k. Let $\boldsymbol{\theta}_k$ and $\boldsymbol{g}_k$ denote the parameter and stochastic gradient at step k, let $\boldsymbol{m}_k$ and $\boldsymbol{v}_k$ denote the momentum and adaptive learning rate at iteration k.

(c) [4 points] Choose the most appropriate algorithm for the following scenario:
We are training a deep neural network on a massive dataset ($n = 10^7$) where we want to apply ℓ_2 regularization (weight decay). Which algorithm is preferred to ensure the weight decay acts as proper ℓ_2 regularization and simultaneously enable fast convergence? 1) Adam, 2) AdamW, 3) SGD. Justify your choice briefly.

Question 3 [10 points]: Neural Network Construction and Analysis

Consider an L-layer feed-forward neural network. The input is denoted by $\boldsymbol{x}$, and the parameters are weights and biases at each layer.

(a) [3 points] Consider a neural network designed for a multi-class classification task. The input dimension is $d = 50$, and there are $K = 10$ classes. The network consists of 2 hidden layers: the first hidden layer has 100 neurons, and the second hidden layer has 20 neurons. Assuming **no bias terms** are included in any layer, calculate the total number of learnable parameters in this network.

(b) [4 points] Consider the input $\boldsymbol{x} = [x_1, x_2]^\top \in \mathbb{R}^2$. Let the activation function for the hidden layer be the ReLU function $\sigma(t) = \max\{0, t\}$, and the output layer activation $h(\cdot)$ be the identity function. Design a one-hidden layer neural network that implements the function $f(\boldsymbol{x}) = |x_1 - x_2|$. Specify the dimensions and specific values of the weights and biases $(\boldsymbol{W}_{(1)}, \boldsymbol{b}_{(1)}, \boldsymbol{w}_{(2)}, b_{(2)})$. Note that the output of the network is given by $f(\boldsymbol{x}) = \boldsymbol{w}_{(2)}^\top\sigma(\boldsymbol{W}_{(1)}\boldsymbol{x} + \boldsymbol{b}_{(1)}) + b_{(2)}$.
Hint: Recall that the absolute value function can be decomposed as $|a| = \max\{0, a\} + \max\{0, -a\}$.

(c) [3 points] Consider a scalar input $x \in \mathbb{R}$. Using the same ReLU activation $\sigma(t)$ for the hidden layer and identity output $h(\cdot)$, design a neural network that implements the "clipping" function defined below:

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x \leq 2 \\ 2 & \text{if } x > 2 \end{cases}$$

Specify the values of the weights and biases $(\boldsymbol{W}_{(1)}, \boldsymbol{b}_{(1)}, \boldsymbol{w}_{(2)}, b_{(2)})$. The output of the network is given by $f(x) = \boldsymbol{w}_{(2)}^\top \sigma(\boldsymbol{W}_{(1)} x + \boldsymbol{b}_{(1)}) + b_{(2)}$.
Hint: You can first try to express the function $f(x)$ in terms of the ReLU activation function $\sigma(\cdot)$.

Question 4 [10 points]: Training vs. Testing

We will discuss the fundamental trade-off between training error and generalization error with respect to model complexity (measured by VC dimension d_{VC}). Refer to Figure 1 below.

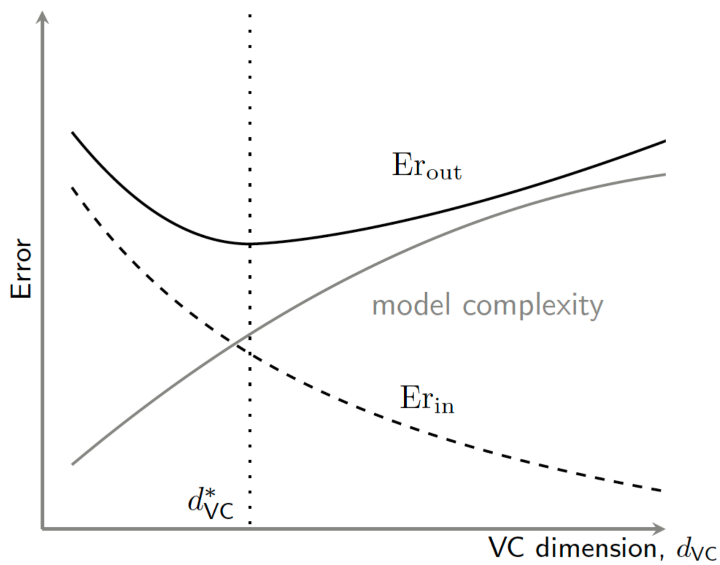

Figure 1: The behavior of in-sample error (Er_{in}) and out-of-sample error (Er_{out}) as a function of the VC dimension d_{VC}.

(a) [5 points] Suppose we collect more training data, increasing the sample size from n to n' (where $n' \gg n$).
  (1) Describe how the curves for Er_{in} and Er_{out} would shift relative to their positions in Figure 1.
  (2) How would this increase in n affect the position of the optimal model complexity d^*_{VC}?

(b) [5 points] (1) Based on the VC generalization bound discussed in our lectures, use big $O(\cdot)$ notation to write down the generalization inequality relating $\text{Er}_{\text{out}}(f)$, $\text{Er}_{\text{in}}(f)$, and the generalization error $\Omega(n, d_{\text{VC}}, \delta)$. (2) Identify which curve in Figure 1 corresponds to this generalization error term (answer by referring to the curve label).

Question 5 [15 points]: Transformers and Large Language Models

The Transformer architecture has become the foundation of modern Large Language Models (LLMs). We will analyze its core component, the Scaled Dot-Product Attention, and its computational properties.

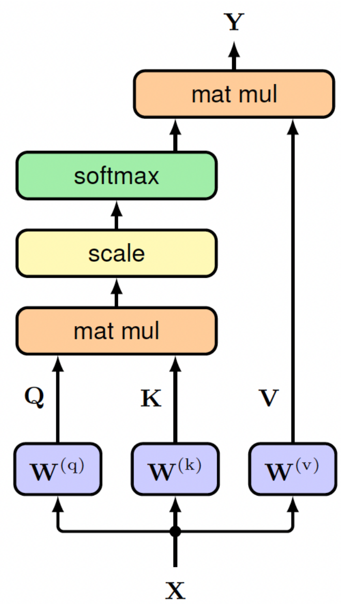

Figure 2: Illustration of the Scaled Dot-Product Attention mechanism.

(a) [3 points] Consider the attention mechanism illustrated in Figure 2. Let the input queries, keys, and values be matrices $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{n \times d}$, where n is the sequence length and d is the dimension.

(1) Write down the mathematical formula for the output matrix $\boldsymbol{Y}$ corresponding to this figure.
(2) Explain the specific role of the "scale" operation shown in the figure.

(b) [3 points] In the context of autoregressive Large Language Models (e.g., GPT, Llama), we must ensure causality during the training process.

(1) Explain why a "Causal Mask" is necessary in the attention mechanism for these models.
(2) Write down the mathematical definition of the causal mask matrix $\boldsymbol{M}^{causal} \in \mathbb{R}^{n \times n}$, i.e., the value of the element M_{ij}^{causal} in the matrix $\boldsymbol{M}^{causal}$.

(c) [3 points] Consider a Multi-Head Attention module with a model dimension $d = 512$ and $H = 8$ heads. We assume the dimension of each head is $d_v = d_k = d/H$.

(1) What are the dimensions of the query weight matrix $\boldsymbol{W}_h^{(q)}$ for a single head?
(2) After concatenating the outputs of all H heads, we apply a linear transformation using matrix $\boldsymbol{W}^{(o)}$. What is the dimension of $\boldsymbol{W}^{(o)}$?

(d) [3 points] Analyze the computational complexity of Self-Attention with sequence length n and model dimension d.

(1) What is the Big-O computational complexity of the Self-Attention module?
(2) What is the storage / memory cost in Big-O notation?

(e) [3 points] (1) The final layer of a Transformer (often called the LM Head) converts the learned features $\boldsymbol{Z}$ into probability distributions for next-token prediction.. Let the final hidden states be $\boldsymbol{Z} \in \mathbb{R}^{(n+1)\times d}$ and the LM head weight matrix be $\boldsymbol{W}_{\text{lm}}$. What must be the dimensions of $\boldsymbol{W}_{\text{lm}}$?

(2) The training of a decoder-only Transformer is formulated as a next-token prediction task using $\boldsymbol{Z}$ and $\boldsymbol{W}_{\text{lm}}$. Identify the specific type of machine learning model (or classifier) implemented by the last layer of the Transformer to perform this prediction.